TD 1

Reminder

$$\mathbf{DPI}_{\text{ppi}} = \frac{\text{height}_{\text{px}}}{\text{height}_{\text{inches}}} = \frac{\text{width}_{\text{px}}}{\text{width}_{\text{inches}}} = \frac{\sqrt{\text{width}_{\text{px}}^2 + \text{height}_{\text{px}}^2}}{\text{diagonal}_{\text{inches}}} = \frac{\text{size}_{\text{px}\times 160}}{\text{DP}}$$

$$\mathbf{DP} = \frac{\text{size}_{\text{px}} \times 160}{\mathbf{DPI}_{\text{ppi}}}$$

Exercice 1

Give the density (**DPI**) and **DP** of the following screens :

1. 1280×720 / D = 4"
2. 800×480 / H = 3" / W = 5"
3. 800×480 / 1000×600 **DP**
4. **DP** = 4px
5. **HD** / 640×360 **DP**
6. **FHD** / D = 4"

Solution Exercice 1

1. 1280×720 / D = 4"

$$\mathbf{DPI} = \frac{\sqrt{1280^2 + 720^2}}{4} = \boxed{367 \text{ ppi}}$$

$$\mathbf{DP}_{\text{width}} = \frac{1280 \times 160}{367} = \boxed{558}$$

$$\mathbf{DP}_{\text{height}} = \frac{720 \times 160}{367} = \boxed{313}$$

2. 800×480 / H = 3" / W = 5"

$$\mathbf{DPI} = \frac{800}{5} = \frac{480}{3} = \frac{\sqrt{1280^2 + 720^2}}{\sqrt{5^2 + 3^2}} = \boxed{160 \text{ ppi}}$$

$$\mathbf{DP}_{\text{width}} = \boxed{800}$$

$$\mathbf{DP}_{\text{height}} = \boxed{480}$$

3. 800×480 / 1000×600 **DP**

$$\mathbf{DPI} = \frac{800 \times 160}{1000} = \frac{480 \times 160}{600} = \boxed{128 \text{ ppi}}$$

4. **DP** $= 4 \times$ px

$$4 \times \text{size}_{\text{px}} = \frac{\text{size}_{\text{px}} \times 160}{\textbf{DPI}}$$

$$\textbf{DPI} = \frac{\cancel{\text{size}_{\text{px}}} \times 160}{4 \times \cancel{\text{size}_{\text{px}}}}$$

$$\textbf{DPI} = \frac{160}{4} = \boxed{40 \text{ ppi}}$$

5. **HD** $= 1280 \times 720$ / 640×360 **DP**

$$\textbf{DPI} = \frac{1280 \times 160}{640} = \frac{720 \times 160}{360} = \boxed{320 \text{ ppi}}$$

6. **FHD** $= 1920 \times 1080$ / D $= 4$"

$$\textbf{DPI} = \frac{\sqrt{1920^2 + 1080^2}}{4} = \boxed{550 \text{ ppi}}$$

$$\textbf{DP}_{\text{width}} = \frac{1920 \times 160}{550} = \boxed{558}$$

$$\textbf{DP}_{\text{height}} = \frac{1080 \times 160}{550} = \boxed{314}$$

Exercice 2

fill the table

Resolution (px)	**DPI**	**Resolution (DP)**	**Size**
480×320	mdpi = 160		
480×320	ldpi = 120		
480×320	hdpi = 240		
640×480	hdpi = 240		
320×240	ldpi = 120		
1920×1080	xxdpi = 480		

Solution Exercice 2

Resolution (px)	**DPI**	**Resolution (DP)**	**Size**
480×320	mdpi = 160	$\text{DP}_{\text{width}} = 480$ dp $\text{DP}_{\text{height}} = 320$ dp	$\frac{\sqrt{480^2 + 320^2}}{160} = 3.6"$
480×320	ldpi = 120	$\text{DP}_{\text{width}} = \frac{480 \times 160}{120} = 640$ dp $\text{DP}_{\text{height}} = \frac{320 \times 160}{120} = 426$ dp	$\frac{\sqrt{480^2 + 320^2}}{120} = 4.8"$
480×320	hdpi = 240	$\text{DP}_{\text{width}} = \frac{480 \times 160}{240} = 320$ dp $\text{DP}_{\text{height}} = \frac{320 \times 160}{240} = 213$ dp	$\frac{\sqrt{480^2 + 320^2}}{240} = 2.4"$
640×480	hdpi = 240	$\text{DP}_{\text{width}} = \frac{640 \times 160}{240} = 426$ dp $\text{DP}_{\text{height}} = \frac{480 \times 160}{240} = 320$ dp	$\frac{\sqrt{640^2 + 480^2}}{240} = 3.3"$
320×240	ldpi = 120	$\text{DP}_{\text{width}} = \frac{320 \times 160}{120} = 426$ dp $\text{DP}_{\text{height}} = \frac{240 \times 160}{120} = 320$ dp	$\frac{\sqrt{320^2 + 240^2}}{120} = 3.3"$
1920×1080	xxhdpi = 480	$\text{DP}_{\text{width}} = \frac{1920 \times 160}{480} = 640$ dp $\text{DP}_{\text{height}} = \frac{1080 \times 160}{480} = 360$ dp	$\frac{\sqrt{1920^2 + 1080^2}}{480} = 4.5"$

Exercice 3

we have an icon of size = 0.5" give its size in px for density lDPI = 120 ppi

Solution Exercice 3

$$\textbf{DPI} = \frac{\text{size}_{\text{px}}}{\text{size}_{\text{inch}}}$$

$$\text{size}_{\text{px}} = \textbf{DPI} \times \text{size}_{\text{inch}} = 120 \times 0.5 = \boxed{60 \text{ px}}$$